

AI CRM Lead Automation

Python 3.11+ local application | Standard-library runtime | Optional OpenAI Responses API | MIT licensed

DELIVERY	ANALYSIS	QUALITY	DEPLOYMENT
Local dashboard + API	Offline or optional OpenAI	66 automated tests	Local-first application

Architecture

INTAKE	ANALYSIS	PERSISTENCE	DELIVERY
Threaded local HTTP server, strict JSON media type, request/message bounds, bearer/session access, CSRF.	Versioned scoring policy or best-effort redacted Structured Outputs; local validation always enforced.	Process lock, duplicate check, transactional state, retention, export/delete, raw message opt-in.	Escaped dashboard, JSON endpoints, date-accurate metrics, event log, formula-safe CSV.

HTTP surface

METHOD	ROUTE	PURPOSE
GET	/ and /api/health	Dashboard and mode-aware health check
POST	/api/intake	Validate, classify, score, and persist one lead
GET	/api/leads and /api/logs	Structured CRM records, metrics, and audit events
GET/DELETE	/api/leads/{id}	Export or delete one contact record
POST	/api/maintenance/purge	Apply configured contact retention
GET	/export/leads.csv	Formula-neutralized spreadsheet handoff

Security and privacy controls

- Loopback-only host validation	- CSV formula neutralization
- Optional strong local bearer/session key	- No secrets or lead text in logs
- Browser CSRF protection	- Raw inquiry retention off by default
- Security and no-cache headers	- 90-day purge plus export/delete by ID

Verification

66 isolated tests cover configuration, safe seed behavior, scoring-policy schema, conservative extraction, strict completed provider output, retries, transactional storage, lifecycle, authentication, CSRF, CSV and HTML safety, and local HTTP behavior. CI enforces 90% overall coverage on Python 3.11-3.13, verifies scoring, and rebuilds both PDFs.

Run locally

```
python3 -m unittest discover -s tests -v      python3 src/seed_data.py --reset      python3 src/app.py
```

Production rollout makes identity and authorization mandatory and adds TLS, rate limits, managed persistence, monitoring, backups, and scheduled lifecycle controls.